Downloading raw CSV Files from TOS

Raw data is downloaded from the Thinkorswim (TOS) scanner as 11 CSV files and stored in **Downloaded CSV Files** folder. Each file is named according to the sector (e.g., Technology.csv, Healthcare.csv).

Follow these steps to update raw data:

1. Log in to Thinkorswim → main window → Scan tab → Stock Hacker
2. Third ribbon:
   Scan in: All Listed Stock (in Category)
   Intersect with: By Industry → Choose an industry → Select All <industry>
   Exclude: All OTC Stocks
3. Use the following filters:
   - Stock: Market cap, $M → min: 250M, no max value is specified
   - Fundamental: Choose any fundamental metric, and leave min and max values empty. This step is done to make sure OTC stocks are excluded as sometimes the Exclude condition doesn't work properly.
4. Make sure required columns are set up in the right order:
   - Use the **Table Schema.xlsx** in the project folder to see the required columns.
   - Click the little settings wheel below the Scan button to customize columns.
5. Click Scan button on the right: It may take a while for data to fully load.
6. At the top right corner, under On Demand: click drop-down menu → Export → To file… → Navigate to **Downloaded CSV Files** folder → Replace the CSV file of the current industry

Download Position Statement

Before running the **exit_signals.py** and **stock_database.py** scripts, download the position statement from TOS and save it in the **Position Statement** folder.

1. Log in to Thinkorswim → main window → Monitor tab → Activity and Positions
2. Position Statement section: make sure all the current positions in the portfolio are displayed → drop-down menu on the right of the section → Export → To file… → Save file to the **Position Statement** folder

Running Python Scripts

Using batch files (for Windows):

1. Before using the .bat files for the first time, change the project folder path:
   - right click a .bat file, choose Edit in Notepad
   - Replace "D:\PROJECTS\Undervalued Stock Scanner V2" with your project folder
   - Save and close the file.
2. Navigate to Batch Processing folder → Double-click a .bat file and follow the Command Prompt instructions to start running the scripts
3. Create an app icon on Desktop (optional): Right-click the .bat file → Send to → Desktop (create shortcut) → Right-click the shortcut on Desktop → Properties → Change Icon → Choose an icon → Apply → OK

Using terminal:

1. Open the project folder terminal: Right-click the project folder → Open in Terminal
2. Install all the required libraries (if not yet installed): Run pip install -r requirements.txt

3. To output stock screening result: Run python -m SCRIPTS.etl
4. To output exit signal detection result and store stock data in my_database.db: Run python -m SCRIPTS.stock_database

Outputs of Python Scripts:

etl.py:

- **Results**\
- **Industry Means**\

exit_signals.py:

- **Exit Signals****Scannable**\
- **Exit Signals****Unscannable**\

stock_database.py:

- **my_database.db** in the project folder
- **stock_data.csv** in the project folder

Using Power BI Dashboard

Follow the these steps when using the .pbix files for the first time:

1. Copy the path of the project folder
2. Open **Undervalued Stock Scanner.pbix** or **Stock Monitor.pbix**
3. Home tab → Click Transform data drop-down → Edit parameters → Click on the FolderPath parameter → Paste the current folder path to replace the default folder path → Click OK
4. Click Apply changes in the popup

Refresh data when you open the dashboards after running Python scripts: Home tab → Click Refresh